Evidence-Grounded Meeting Intelligence with Local Speech Recognition and State-Preserving Reanalysis

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Abstract

Recorded meetings contain decisions, commitments, and unresolved questions that are difficult to retrieve from lengthy transcripts. Although language models can organize this information, practical meeting analysis systems also require traceable evidence, consistent report updates, and preservation of human corrections. This paper presents Meeting Analyzer, an integrated system combining local speech recognition with structured language-model extraction and evidence-aware state reconciliation. Uploaded recordings are normalized using FFmpeg and transcribed using Faster-Whisper. A schema-constrained semantic analysis stage produces summaries, action items, decisions, questions, topics, and entities. Extracted evidence is checked against the transcript and associated with source segments to support audio navigation. Versioned input and configuration manifests support report comparison, while reconciliation preserves eligible prior items and user edits. Preliminary evaluation uses nine recorded transcription executions over three overlapping excerpts from one recording and five archived semantic analyses of one controlled fixture. Mean CPU transcription times are 2.315, 5.715, and 11.562 seconds for 30-, 60-, and 120-second excerpts, respectively. Action, decision, and question identifiers remain consistent across the archived analyses, whereas topic agreement varies. These findings demonstrate local transcription efficiency and category-dependent reproducibility in limited cases; they do not establish extraction accuracy or superiority over competing systems.

Index Terms: Meeting analysis, automatic speech recognition, large language models, evidence grounding, action-item extraction, state reconciliation, meeting summarization.

I. INTRODUCTION

Meetings support collaborative planning, technical discussion, and organizational decision-making. Their outcomes are distributed across conversational turns, making it difficult to recover responsibilities, decisions, and unresolved issues from recordings. Manual minutes require sustained attention, while a complete transcript often remains too lengthy for efficient review.

Automatic speech recognition provides a searchable textual representation of recorded conversations. Whisper demonstrates the applicability of large-scale weak supervision to speech recognition [1]. Nevertheless, transcription alone does not organize meeting content into operational outcomes. A useful meeting analysis system must distinguish proposed activities from accepted commitments and avoid assigning owners or deadlines that the conversation does not establish.

Language models can transform transcripts into structured reports, but their integration introduces engineering challenges. Generated statements may lack sufficient support, repeated analyses may change report contents, and newly generated outputs may overwrite human corrections. Long transcripts also require controlled processing to avoid silently excluding relevant content.

This work combines local speech recognition, structured semantic extraction, transcript evidence validation, and state-preserving report updates. Its contributions are an integrated recording-to-report workflow, evidence-linked navigation, versioned reconciliation that distinguishes generated fields from user corrections, and a qualified preliminary evaluation. The contribution concerns system integration and auditable report management, rather than a new speech recognition or language-model architecture.

II. RELATED WORK

A. Automatic Speech Recognition

Whisper uses large-scale weak supervision for speech recognition and related tasks [1]. Meeting Analyzer uses Faster-Whisper as its transcription implementation. Recognition errors in names, dates, or task descriptions can propagate into extracted outcomes. This work evaluates local transcription runtime under one configuration; it does not evaluate recognition accuracy or modify Whisper.

B. Meeting Summarization

QMSum formalizes query-based meeting summarization, in which systems locate and summarize transcript spans relevant to a user query [2]. Meeting Analyzer instead produces several structured outcome categories and manages them as persistent, user-editable records. Its focus includes evidence navigation and preservation of report state during subsequent analyses.

C. Evaluation and Positioning

ROUGE measures overlap between generated summaries and reference summaries [3]. Such metrics support content evaluation when human references exist, but overlap alone does not establish correct ownership, deadlines, or decisions. This system therefore requires evaluation of transcription, extraction, summaries, evidence, runtime, and stability. Whether its state-management mechanisms improve accuracy or productivity remains an empirical question.

III. PROJECT DETAILS AND ARCHITECTURE

Meeting Analyzer accepts recorded audio or video and produces a timestamped transcript, summary, key takeaway, actions, decisions, questions, topics, entities, and an exportable report. Missing information remains unspecified. The interface supports inspection of evidence associated with extracted outcomes.

Component	Implementation	Role
Client	React, Vite, Tailwind CSS, Recharts	Upload, report interaction, charts
API	FastAPI	Upload, analysis, retrieval, export
Media and ASR	FFmpeg, Faster-Whisper	Audio normalization and local transcription
Semantic analysis	Groq-compatible structured inference	Meeting information extraction
Validation	Pydantic and application validators	Canonical structure and evidence checks
Reconciliation	Custom Python service	Item matching and edit preservation
Storage and export	Supabase, local recovery files, ReportLab	Persistence and PDF reports

The repository also contains classical NLP and summarization modules, including spaCy-based processing, classification, and transformer summarization. These are separate legacy or experimental services, not sequential stages of the current canonical semantic report pipeline.

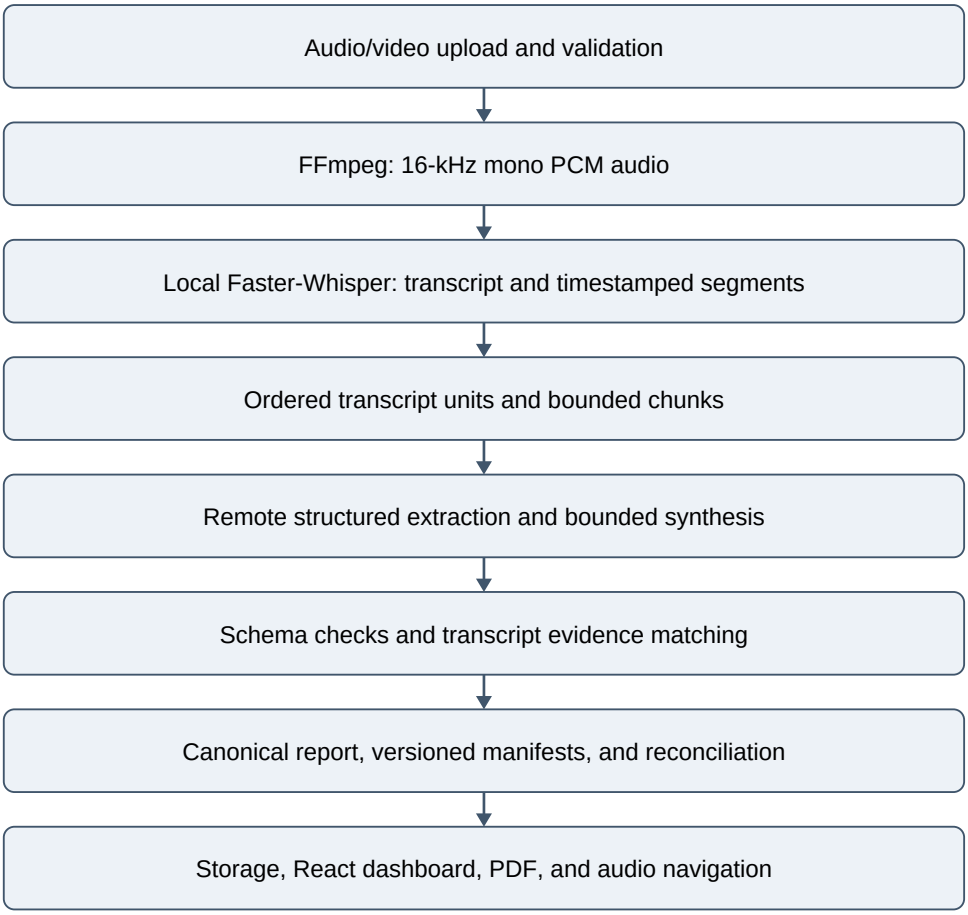


Fig. 1. Current recording-to-report architecture. Previous reports and user edits are additional reconciliation inputs.

Speech recognition executes locally, but the configured semantic analysis stage sends transcript content to a remote provider. The current workflow is therefore not fully offline.

IV. METHODOLOGY

A. Audio Processing and Transcription

FFmpeg converts recordings into mono, 16-kHz PCM WAV audio. Faster-Whisper represents the transcript as T = {s1, ..., sn}, where each segment contains its identifier, start time, end time, and recognized text. Voice activity detection is enabled, and segment timestamps are retained. The current transcription service does not perform speaker diarization.

B. Transcript Partitioning

Ordered transcript units are divided when the configured request budget is exceeded. Each chunk satisfies bytes(serialize(chunk)) <= B, where B is the configured payload ceiling. One turn of overlap is included where the budget permits; oversized turns are split. Multiple chunk reports are combined through bounded synthesis. A byte budget controls payload size but is not identical to a token count.

C. Structured Extraction

The semantic model produces summaries, actions, decisions, questions, follow-ups, topics, entities, and supporting evidence. An action is represented by its task description, owner, deadline, evidence quote, and source segment identifiers. Owners and deadlines remain null when unavailable. Structured output constrains representation; subsequent validators check individual fields.

D. Evidence Validation

Evidence quotes are matched against normalized transcript text and cited source spans. Unambiguous matches support navigation to the first supporting segment timestamp. Ambiguous matches should not be treated as precise temporal locations. Quote matching establishes textual occurrence, not logical entailment or transcription

correctness.

E. Versioned Analysis Identity

Input and configuration manifests use SHA-256 digests of serialized data. Configuration records include model identity, schema and prompt hashes, processing budgets, and chunking policy. Compatibility checks compare manifests and version fields before retention rules are applied. Hashes support provenance; they do not guarantee identical provider outputs.

F. State Reconciliation

Item identifiers derive from the meeting, category, supporting segments, and a text discriminator when necessary. Prior and new items are compared using evidence overlap and content checks. Matched items preserve user corrections. Eligible prior items may be retained when analysis settings are compatible and evidence remains valid. Changed or unsupported items may be removed, with edited records retained for manual review. Topics use current output rather than accumulating prior topic labels.

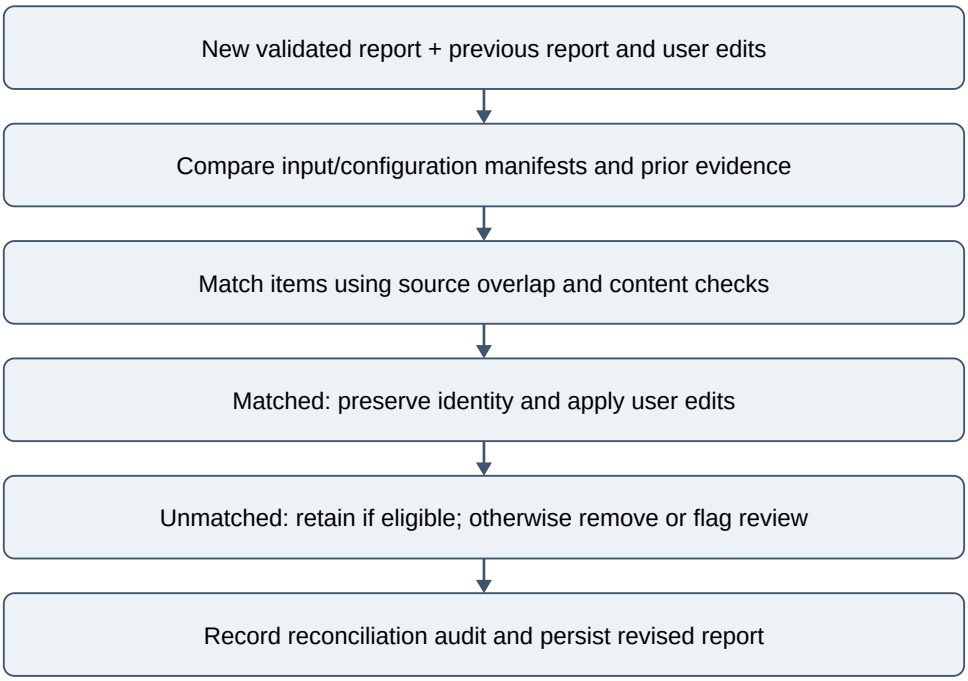


Fig. 2. Reconciliation procedure. The matched and unmatched rows describe alternative cases; topics follow a separate current-output policy.

Retention can improve continuity but may also preserve an incorrect interpretation supported by a real quotation. Reconciliation must be evaluated separately from extraction correctness.

V. EXPERIMENTAL SETUP

A. Objectives and Data

The available evidence addresses local transcription runtime and item-identifier reproducibility. The timing benchmark uses 30-, 60-, and 120-second prefixes of one approximately 550-second recording, with three timed runs per prefix. The semantic case study uses five archived provider analyses of one 47-second English fixture; Runs 2-5 are compared with Run 1.

Parameter	Recorded configuration
Environment	Windows 11; Python 3.12.10
Processor	AMD64 Family 25 Model 117; 16 logical processors
ASR	Faster-Whisper base; CPU; INT8; beam size 5
Speech processing	VAD enabled; minimum silence 500 ms; automatic language detection

Parameter	Recorded configuration
Timing protocol	One 30-second warm-up; nine timed runs; randomized order, seed 20261025
Archived semantic model	openai/gpt-oss-120b via Groq; temperature 0; content-derived seed

Timing includes decoding, VAD, language detection, complete consumption of the transcription iterator, and output construction. It excludes model loading, clip preparation, semantic analysis, and persistence. Recorded model loading took approximately 0.483 seconds. The semantic experiment is an archived case study, not fresh remote inference.

B. Metrics

Real-time factor (RTF) = processing time / audio duration. An RTF below one indicates faster-than-real-time execution of the measured stage. Variability is reported using sample standard deviation.

Jaccard agreement J(A,B) = |intersection(A,B)| / |union(A,B)| compares sets of item identifiers. Empty follow-up sets are excluded. Identifier agreement does not establish identical wording, factual accuracy, or complete extraction.

VI. RESULTS AND ANALYSIS

A. Transcription Runtime

Audio	Runs	Mean time +/- sample SD	Mean RTF
30 s	3	2.315 +/- 0.483 s	0.0772
60 s	3	5.715 +/- 0.563 s	0.0952
120 s	3	11.562 +/- 0.537 s	0.0963

All measured mean real-time factors are below 0.10. Warm transcription required less than one-tenth of the corresponding audio duration for these excerpts. This supports local transcription feasibility on the recorded machine, but does not establish full application latency.

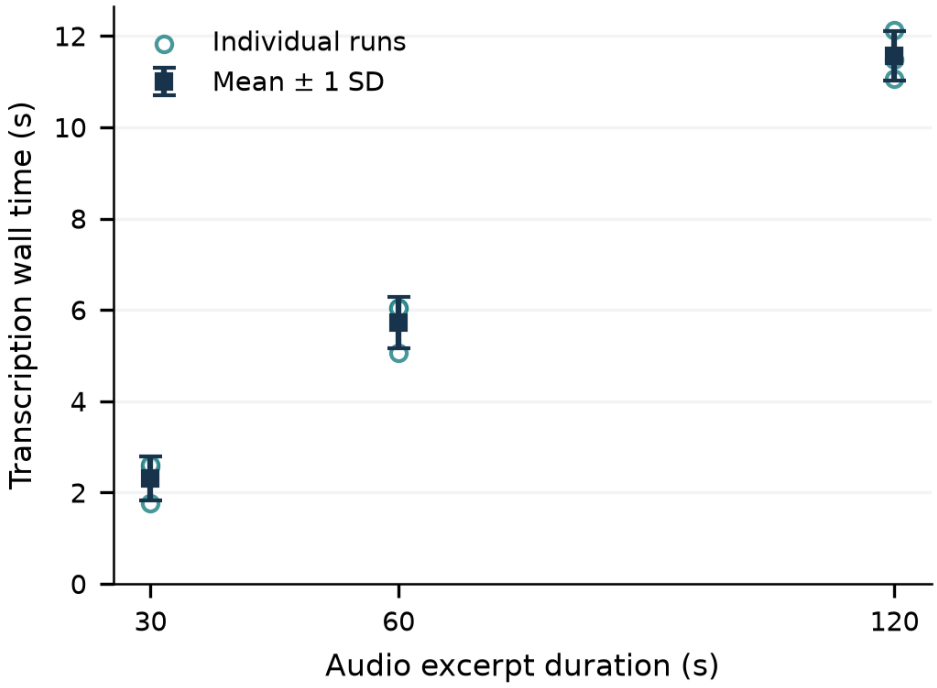


Fig. 3. Warm CPU INT8 transcription latency for overlapping prefixes of one recording. Points show three timed runs per duration; summary markers and whiskers show mean and sample standard deviation. Loading and downstream analysis are excluded.

The excerpts overlap and are correlated observations from one recording. Technical repetitions measure local execution variability, not general performance across independent meetings.

B. Reanalysis Agreement

Category	Run 2 vs. 1	Run 3 vs. 1	Run 4 vs. 1	Run 5 vs. 1
Actions	1.000	1.000	1.000	1.000
Decisions	1.000	1.000	1.000	1.000
Questions	1.000	1.000	1.000	1.000
Topics	0.000	1.000	0.667	1.000

Fresh-candidate and persisted-stage values are identical. Every run contains two actions, one decision, and one question; topic counts vary from two to five. Since candidate outputs already have perfect agreement in the first three categories, the experiment does not demonstrate an improvement caused by reconciliation.

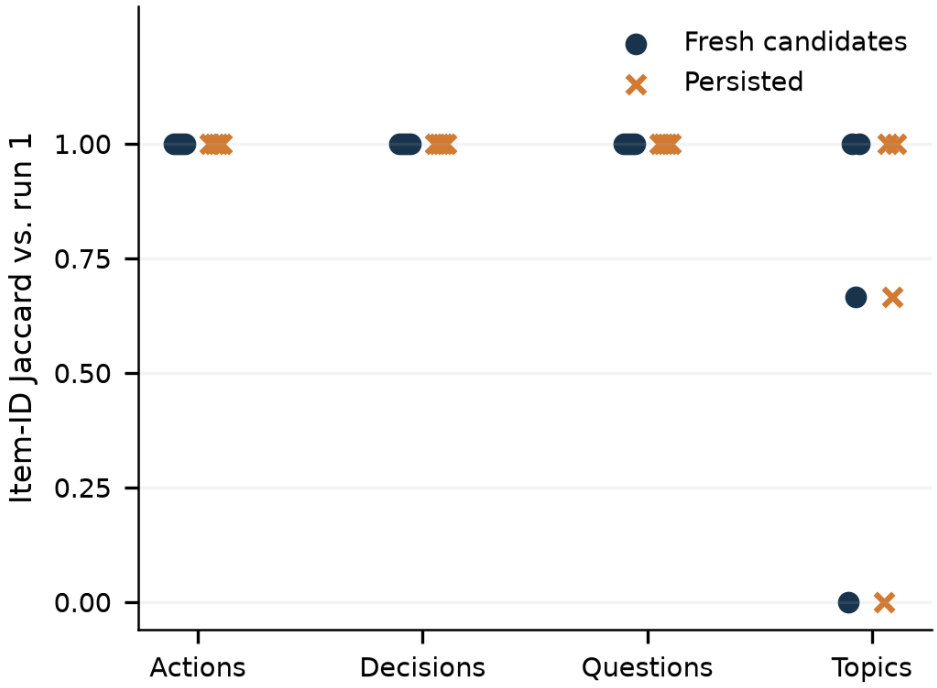


Fig. 4. Item-ID agreement relative to the first of five archived analyses. Candidate and persisted results coincide; topic output varies.

Topic variability demonstrates that the full report is not uniformly reproducible. Stable identifiers in selected categories should not be described as universal deterministic generation.

C. Accuracy and Summary Quality

The supplied evaluation lacks independently annotated references sufficient to report word error rate, extraction precision/recall/F1, owner or deadline accuracy, ROUGE scores, or hallucination rate. Training examples and software fixtures are not substitutes for an independent test corpus.

VII. COMPARISON WITH OTHER APPROACHES

The following comparison concerns deliberately defined baseline designs, not named commercial products. It identifies implemented capabilities rather than measured superiority.

Dimension	Transcript only	Single-pass summary	Proposed system
Output	Transcript	Generated summary	Structured report
Outcome categories	Absent	Prompt-dependent	Explicit categories
Evidence fields	Not applicable	Not required	Included and validated
Navigation	Transcript timestamps	Not specified	Evidence-linked segments

Dimension	Transcript only	Single-pass summary	Proposed system
Reanalysis	Not applicable	Replace output	Reconcile prior state
Human edits	Separate feature	Not specified	Explicit preservation
Provenance	Not specified	Not specified	Versioned manifests
Relative accuracy	Not evaluated	Not evaluated	Not evaluated

The practical advantages are traceability, explicit representation of meeting outcomes, continuity across reanalysis, and inspectable processing state. These features support an auditable workflow. They do not prove higher accuracy, lower cost, or faster operation than competing systems. Comparative superiority requires matched evaluation on held-out meetings.

VIII. KEY FINDINGS AND LIMITATIONS

The recorded CPU benchmark shows warm transcription faster than real time for all tested excerpts. The implementation supports evidence-linked reporting. Actions, decisions, and questions retain stable identifiers in one archived fixture, whereas topics vary. Candidate and persisted agreement are identical, so reconciliation benefit is not isolated.

The evaluation includes one recording for timing and one semantic fixture. Overlapping clips and repeated executions are not independent meetings. Evidence matching validates occurrence rather than entailment. ASR errors can propagate downstream, and the absence of diarization limits speaker attribution. Remote semantic inference prevents a fully offline workflow. No quantified cost or privacy comparison is available. Retention may preserve prior mistakes, requiring evaluation of continuity alongside correctness.

IX. CONCLUSION

Meeting Analyzer integrates local speech recognition, structured semantic extraction, and state-preserving reconciliation. The architecture connects meeting outcomes to transcript evidence and preserves eligible user corrections during reanalysis. Preliminary mean transcription times range from 2.315 to 11.562 seconds for 30-120-second excerpts on the recorded CPU configuration. An archived case study shows stable action, decision, and question identifiers alongside variable topic output. These findings support implementation feasibility while leaving general extraction quality and comparative effectiveness unresolved.

X. FUTURE WORK

Independent evaluation: Create adjudicated references for transcripts, outcomes, and evidence across multiple held-out meetings.

Baseline and ablation experiments: Compare rules-based extraction, unconstrained semantic extraction, evidence validation, and the complete reconciliation pipeline on identical inputs.

Speaker diarization: Evaluate speaker-attributed transcripts and task ownership.

Semantic evidence assessment: Test whether quotations entail generated claims rather than merely occurring in source text.

Long and multilingual meetings: Measure coverage, cross-chunk dependencies, latency, failures, language effects, and code switching.

Local inference and user studies: Evaluate fully local semantic models, memory and runtime requirements, correction effort, and verification time.

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MANUSCRIPT PREPARATION NOTE

This PDF is a readable manuscript draft with IEEE-style section organization, not a claim of compliance with a specific conference template. Replace author placeholders and apply the target venue layout and page limit. Figures 3-4 use existing recorded experimental data. No accuracy or competitor-comparison graph is included because the required measurements are unavailable.